Quantum Computing and Quantum Machine Learning

Morten Hjorth-Jensen[1]

Department of Physics, University of Oslo[1]

March 5, 2025

Plans for the week of March 3-7

1. Reminder on basics of the VQE method and how to perform measurements for the simpler one- and two-qubit Hamiltonians
2. How to perform measurements
3. Simulating efficiently Hamiltonians on quantum computers with the VQE method and gradient descent to optimize the state function ansatz
4. Introducing the Lipkin model
5. Work on project 1

Readings

1. For the discussion of one-qubit, two-qubit and other gates, sections 2.6-2.11 and 3.1-3.4 of Hundt's book **Quantum Computing for Programmers**, contain most of the relevant information. We will repeat some of the these properties during the first part of the lecture.
2. The VQE algorithm is discussed in Hundt's section 6.11
3. See also the VQE review article by Tilly et al.

Reminder from last week: Variational Quantum Eigensolver

One initial algorithm to estimate the eigenenergies of a quantum Hamiltonian was quantum phase estimation. In it, one encodes the eigenenergies, one binary bit at a time (up to n bits), into the complex phases of the quantum states of the Hilbert space for n qubits. It does this by applying powers of controlled unitary evolution operators to a quantum state that can be expanded in terms of the Hamiltonian's eigenvectors of interest. The eigenenergies are encoded into the complex phases in such a way that taking the inverse quantum Fourier transformation (see Hundt sections 6.1-6.2) of the states into which the eigen-energies are encoded results in a measurement probability distribution that has peaks around the bit strings that represent a binary fraction which corresponds to the eigen-energies of the quantum state acted upon by the controlled unitary operators.

The VQE

While quantum phase estimation (QPE) is provably efficient, non-hybrid, and non-variational, the number of qubits and length of circuits required is too great for our NISQ era quantum computers. Thus, QPE is only efficiently applicable to large, fault-tolerant quantum computers that likely won't exist in the near, but the far future.

Therefore, a different algorithm for finding the eigen-energies of a quantum Hamiltonian was put forth in 2014 called the variational quantum eigensolver, commonly referred to as VQE. The algorithm is hybrid, meaning that it requires the use of both a quantum computer and a classical computer. It is also variational, meaning that it relies, ultimately, on solving an optimization problem by varying parameters and thus is not as deterministic as QPE. The variational quantum eigensolver is based on the variational principle:

Rayleigh-Ritz variational principle

Out starting point is the Rayleigh-Ritz variational principle states that for a given Hamiltonian H, the expectation value of a trial state or just ansatz $|\psi\rangle$ puts a lower bound on the ground state energy E_0.

$$\frac{\langle\psi|\mathcal{H}|\psi\rangle}{\langle\psi|\psi\rangle} \geq E_0.$$

The ansatz

The ansatz is typically chosen to be a parameterized superposition of basis states that can be varied to improve the energy estimate, $|\psi\rangle \equiv |\psi(\boldsymbol{\theta})\rangle$ where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_M)$ are the M optimization parameters.

Expectation value of Hamiltonian and the variational principle

The expectation value of a Hamiltonian $\mathcal{H}$ in a state $|\psi(\theta)\rangle$ parameterized by a set of angles θ, is always greater than or equal to the minimum eigen-energy E_0. To see this, let $|n\rangle$ be the eigenstates of $\mathcal{H}$, that is

$$\mathcal{H}|n\rangle = E_n|n\rangle.$$

Expanding in the eigenstates

We can then expand our state $|\psi(\theta)\rangle$ in terms of the eigenstates

$$|\psi(\theta)\rangle = \sum_n c_n|n\rangle,$$

and insert this in the expression for the expectation value (note that we drop the denominator in the Rayleigh-Ritz ratio)

$$\langle\psi(\theta)|\mathcal{H}|\psi(\theta)\rangle = \sum_{nm} c_m^* c_n\langle m|\mathcal{H}|n\rangle = \sum_{nm} c_m^* c_n E_n\langle m|n\rangle = \sum_{nm} \delta_{nm} c_m^* c_n E_n$$

which implies that we can minimize over the set of angles θ and arrive at the ground state energy E_0

$$\min_{\theta} \langle\psi(\theta)|\mathcal{H}|\psi(\theta)\rangle = E_0.$$

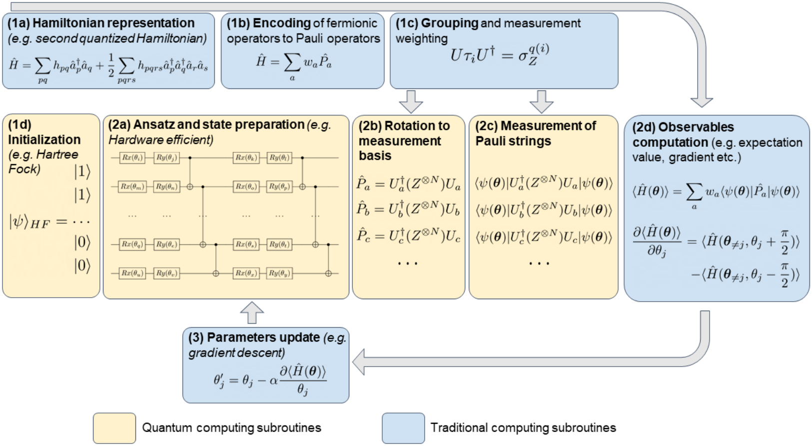
Basic steps of the VQE algorithm

Using this fact, the VQE algorithm can be broken down into the following steps

1. Prepare the variational state $|\psi(\theta)\rangle$ on a quantum computer.
2. Measure this circuit in various bases and send these measurements to a classical computer
3. The classical computer post-processes the measurement data to compute the expectation value $\langle\psi(\theta)|\mathcal{H}|\psi(\theta)\rangle$
4. The classical computer varies the parameters θ according to a classical minimization algorithm and sends them back to the quantum computer which runs step 1 again.

This loop continues until the classical optimization algorithm terminates which results in a set of angles θ_{min} that characterize the ground state $|\phi(\theta_{\text{min}})\rangle$ and an estimate for the ground state energy $\langle\psi(\theta_{\text{min}})|\mathcal{H}|\psi(\theta_{\text{min}})\rangle$.

VQE overview



Why do we measure on one qubit? First consideration

In quantum computing, measurements are typically performed on one qubit at a time due to a combination of theoretical, practical, and algorithmic considerations:

Algorithmic Requirements:

1. Adaptive Processing: Many quantum algorithms, such as quantum teleportation or error correction, require mid-circuit measurements. The outcomes determine subsequent operations, necessitating sequential measurements to adapt the circuit dynamically.
2. Partial Information Extraction: Algorithms often need only specific qubits' results (e.g., in Shor's algorithm), making full-system measurements unnecessary.

Why do we measure on one qubit? Second consideration

Quantum Mechanical Principles:

1. Collapse and Entanglement: Measuring a qubit collapses its state, potentially affecting entangled qubits. Sequential measurements allow controlled extraction of information while managing entanglement.
2. Measurement Basis: Most algorithms use the computational basis (individual qubit measurements). Joint measurements in entangled bases are possible but require complex setups and are not always needed.

Why do we measure on one qubit? Third consideration

Practical Hardware Limitations:

1. Crosstalk and Noise: Simultaneous measurements risk disturbing neighboring qubits due to hardware imperfections, especially in noisy intermediate-scale quantum (NISQ) devices.
2. Readout Constraints: Physical implementations (e.g., superconducting qubits) may have limited readout bandwidth, forcing sequential measurements.

Why do we measure on one qubit? Fourth consideration

Resource Management:

1. Qubit Reuse: Ancilla qubits (e.g., in error correction) are measured, reset, and reused, requiring sequential handling to avoid disrupting computational qubits.

Conclusion:

While joint measurements are theoretically possible, the dominant practice of measuring one qubit at a time stems from algorithmic adaptability, hardware limitations, and the need to minimize quantum state disturbance. This approach balances efficiency, practicality, and the constraints of current quantum systems.

The code for the one qubit case

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns; sns.set_theme(font_scale=1.5)
from tqdm import tqdm

sigma_x = np.array([[0, 1], [1, 0]])
sigma_y = np.array([[0, -1j], [1j, 0]])
sigma_z = np.array([[1, 0], [0, -1]])
I = np.eye(2)

def Hamiltonian(lmb):
    E1 = 0
    E2 = 4
    V11 = 3
    V22 = -3
    V12 = 0.2
    V21 = 0.2

    eps = (E1 + E2) / 2
    omega = (E1 - E2) / 2
    c = (V11 + V22) / 2
    omega_z = (V11 - V22) / 2
    omega_x = V12

    H0 = eps * I + omega * sigma_z
    H1 = c * I + omega_z * sigma_z + omega_x * sigma_x
    return H0 + lmb * H1
```

Two-qubit Hamiltonian

We end this review from last week with a discussion on how to rewrite the two-qubit Hamiltonian rom last week (and project 1)

$$\mathcal{H} = \begin{bmatrix} \epsilon_1 + V_z & 0 & 0 & V_x \\ 0 & \epsilon_2 - V_z & V_x & 0 \\ 0 & H_x & \epsilon_3 - V_z & 0 \\ H_x & 0 & 0 & \epsilon_4 + V_z \end{bmatrix}.$$

This Hamiltonian can be rewritten in terms of various one-qubit matrices.

Definitions

We define

$$\epsilon_{II} = \frac{\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4}{4},$$

$$\epsilon_{ZI} = \frac{\epsilon_1 + \epsilon_2 - \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{IZ} = \frac{\epsilon_1 - \epsilon_2 + \epsilon_3 - \epsilon_4}{4},$$

$$\epsilon_{ZZ} = \frac{\epsilon_1 - \epsilon_2 - \epsilon_3 + \epsilon_4}{4}.$$

The Hamiltonian in terms of Pauli-$\boldsymbol{X}$ and Pauli-$\boldsymbol{Z}$ matrices

With these definitions we can rewrite our two-qubit Hamiltonian as

$$\mathcal{H} = \mathcal{H}_0 + \mathcal{H}_I$$

with

$$\mathcal{H}_0 = \epsilon_{II} \boldsymbol{I} \otimes \boldsymbol{I} + \epsilon_{ZI} \boldsymbol{Z} \otimes \boldsymbol{I} + \epsilon_{IZ} \boldsymbol{I} \otimes \boldsymbol{Z} + \epsilon_{ZZ} \boldsymbol{Z} \otimes \boldsymbol{Z},$$

and

$$\mathcal{H}_I = V_z \boldsymbol{Z} \otimes \boldsymbol{Z} + V_x \boldsymbol{X} \otimes \boldsymbol{X}.$$

How do we perform measurements?

The above tensor products have to be rewritten in terms of specific transformations so that we can perform the measurements in the basis of the Pauli-$\boldsymbol{Z}$ matrices. As we discussed earlier, we need to find a transformation of the form

$$\mathcal{P} = \boldsymbol{U}^{\dagger}\boldsymbol{M}\boldsymbol{U},$$

where $\mathcal{P}$ represents some combination of the Pauli matrices and the identity matrix, $\boldsymbol{U}$ is a unitary matrix and $\boldsymbol{M}$ represents the gate/matrix which performs the measurements, often represented by a Pauli-$\boldsymbol{Z}$ gate/matrix.

Rewriting our strings of Pauli matrices

As discussed last week and reviewed above, to perform a measurement, it is often useful to rewrite the string of Pauli matrices in a specific order or to simplify the expression. Consider a string of Pauli matrices of the form:

$$P = \sigma_{i_1} \otimes \sigma_{i_2} \otimes \cdots \otimes \sigma_{i_n},$$

where each σ_{i_k} is one of the Pauli matrices $\boldsymbol{X}$, $\boldsymbol{Y}$, $\boldsymbol{Z}$, or the identity matrix $\boldsymbol{I}$.

Rewriting the string of matrices

To rewrite this string for measurement purposes, follow these steps:

Commute and reorder:

Use the commutation relations of the Pauli matrices to reorder the string. The commutation relations are:

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k,$$

where ϵ_{ijk} is the Levi-Civita symbol. This allows you to move certain Pauli matrices to the left or right in the string.

Simplify using identities:

Use the fact that $\sigma_i^2 = I$ and $\sigma_i\sigma_j = -\sigma_j\sigma_i$ for $i \neq j$ to simplify the expression. For example:

$$\boldsymbol{XY} = -\boldsymbol{YX} = i\boldsymbol{X}.$$

More rewriting tricks

Group terms:

Group terms that are easier to measure together. For example, if you have a term like $\boldsymbol{X} \otimes \boldsymbol{X}$, you can measure both qubits in the $\boldsymbol{X}$-basis simultaneously.

Diagonalize if necessary:

If the final expression is not diagonal, you may need to apply a unitary transformation to diagonalize it before measurement. For example,to measure $\boldsymbol{X}$, you can apply the Hadamard gate $\boldsymbol{H}$ to transform it into $\boldsymbol{X}$:

$$H\boldsymbol{X}H = \boldsymbol{Z}.$$

Example

A simple example is

$$\boldsymbol{P} = \boldsymbol{X}\boldsymbol{Y} \otimes \boldsymbol{Z}.$$

To rewrite this for measurement we use the commutation relations to reorder the terms if needed. Then we eimplify using identities:

$$\boldsymbol{X}\boldsymbol{Y} = i\boldsymbol{Z}.$$

We obtain then

$$\boldsymbol{P} = (i\boldsymbol{Z}) \otimes \boldsymbol{Z} = i(\boldsymbol{Z} \otimes \boldsymbol{Z}).$$

The $\boldsymbol{Z} \otimes \boldsymbol{I}$ term

The explicit matrix is

$$\boldsymbol{Z} \otimes \boldsymbol{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

When we perform a measurement on the first qubit, we see that this matrix gives us the correct eigenvalues for the first qubit (but not for the second one). To see this multiply the above matrix with our computational basis states, that is the states $|00\rangle = |0\rangle \times |0\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.

This process is referred to in the language of Pauli measurements as **measuring Pauli-Z** and is entirely equivalent to performing a computational basis measurement.

The specific eigenvalues

Multiplying with the $|00\rangle$ state we get

$$\boldsymbol{Z} \otimes \boldsymbol{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

and

$$\boldsymbol{Z} \otimes \boldsymbol{I} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

We don't get the correct eigenvalues if we perform the measurement on the second qubit!

The $\boldsymbol{I} \otimes \boldsymbol{Z}$ term

Our strategy is to rewrite all the strings of Pauli operators via specific unitary transformations in order to obtain a final operator $\boldsymbol{P} = \boldsymbol{Z}_1 \otimes \boldsymbol{I}_2 \otimes \boldsymbol{I}_3 \otimes \cdots \otimes \boldsymbol{I}_N$, where the subscripts indicate the various qubits. We can then perform the measurement on the first qubit only.
We can rewrite $\boldsymbol{I} \otimes \boldsymbol{Z}$ via the SWAP gate

$$\mathrm{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

More on the $\boldsymbol{I} \otimes \boldsymbol{Z}$ term

We have then that

$$\boldsymbol{P} = \text{SWAP}^{\dagger}(\boldsymbol{I} \otimes \boldsymbol{Z})\text{SWAP} = \boldsymbol{Z} \otimes \boldsymbol{I}.$$

We note that the original $\boldsymbol{I} \otimes \boldsymbol{Z}$ does not give the correct eigenvalues when measured on the first qubit. Try this as an exercise.

The $\boldsymbol{Z} \otimes \boldsymbol{Z}$ term

Thus the tensor products of two Pauli-$\boldsymbol{Z}$ operators forms a matrix composed of two spaces consisting of $+1$ and -1 as eigenvalues. As with the single-qubit case, both constitute a half-space, meaning that half of the accessible vector space belongs to the eigenspace with eigenvalue $+1$ and the remaining half to the eigenspace with eigenvalue -1.

This term gives the correct eigenvalue when operating on the first qubit. In principle thus we don't need to rewrite string of operators. However, as discussed below as well, let us rewrite it via a unitary transformation in order to have $\boldsymbol{P} = \boldsymbol{Z} \otimes \boldsymbol{I}$. To do so, consider the transformation

$$\boldsymbol{P} = CX_{10}^{\dagger}(\boldsymbol{Z} \otimes \boldsymbol{Z})CX_{10},$$

where we have

$$\mathrm{CX}_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Performing the transformation

Performing the operation gives

$$\boldsymbol{P} = CX_{10}^{\dagger}(\boldsymbol{Z} \otimes \boldsymbol{Z})CX_{10} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = \boldsymbol{Z} \otimes \boldsymbol{I},$$

which allows us to transform the original tensor product $\boldsymbol{Z} \otimes \boldsymbol{Z}$ into a matrix where we perform the measurement in the basis of the first qubit.
To see this, act with $\boldsymbol{P}$ on the states $|00\rangle = |0\rangle \times |0\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$.

Transformations

Any unitary transformation of such matrices also describes two half-spaces labeled with eigenvalues. For example

$$\boldsymbol{X} \otimes \boldsymbol{X} = \boldsymbol{H} \otimes \boldsymbol{H}(\boldsymbol{Z} \otimes \boldsymbol{Z})\boldsymbol{H} \otimes \boldsymbol{H}, ,$$

follows from from the identity that $\boldsymbol{Z} = \boldsymbol{HXH}$. Similar to the one-qubit case, all two-qubit Pauli-measurements can be written in terms of unitary transformations $\boldsymbol{U}^{\dagger}(\boldsymbol{Z} \otimes \boldsymbol{I})\boldsymbol{U}$ with $\boldsymbol{U}$ being 4×4 unitary matrices.

More terms

Let us look at the $\boldsymbol{X} \otimes \boldsymbol{X}$ term in the simpler two-qubit Hamiltonian.
This term can be rewritten as

$$\boldsymbol{P} = (CX_{10}(\boldsymbol{H} \otimes \boldsymbol{H}))^{\dagger}(\boldsymbol{X} \otimes \boldsymbol{X})(CX_{10}(\boldsymbol{H} \otimes \boldsymbol{H})),$$

which results in

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}.$$

We recognize this result as our $\boldsymbol{Z} \otimes \boldsymbol{I}$ tensor product.

Explicit expressions

In order to perform our measurements for our simple two-qubit Hamiltonian we need the following unitary transformations $\boldsymbol{U}$

$$
\begin{array}{rl}
\boldsymbol{Z} \otimes \boldsymbol{I} & \boldsymbol{U} = \boldsymbol{I} \otimes \boldsymbol{I} \\
\boldsymbol{I} \otimes \boldsymbol{Z} & \boldsymbol{U} = \mathrm{SWAP} \\
\boldsymbol{Z} \otimes \boldsymbol{Z} & \boldsymbol{U} = CX_{10} \\
\boldsymbol{X} \otimes \boldsymbol{X} & \boldsymbol{U} = CX_{10}(\boldsymbol{H} \otimes \boldsymbol{H})
\end{array}
$$

These are the gates that are relevant for the simpler two-qubit Hamiltonian of project 1.

More complete list and derivations of expressions for strings of operators

For a two qubit system we list here the possible transformations

$$
\begin{array}{ll}
\boldsymbol{Y} \otimes \boldsymbol{I} & \boldsymbol{U} = \boldsymbol{HS}^{\dagger} \otimes \boldsymbol{I} \\
\boldsymbol{I} \otimes \boldsymbol{X} & \boldsymbol{U} = (\boldsymbol{H} \otimes \boldsymbol{I})\mathrm{SWAP} \\
\boldsymbol{I} \otimes \boldsymbol{X} & \boldsymbol{U} = (\boldsymbol{H} \otimes \boldsymbol{I})\mathrm{SWAP} \\
\boldsymbol{I} \otimes \boldsymbol{X} & \boldsymbol{U} = (\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{I})\mathrm{SWAP} \\
\boldsymbol{X} \otimes \boldsymbol{Z} & \boldsymbol{U} = CX_{10}(\boldsymbol{H} \otimes \boldsymbol{I}) \\
\boldsymbol{Y} \otimes \boldsymbol{Z} & \boldsymbol{U} = CX_{10}(\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{I}) \\
\boldsymbol{Z} \otimes \boldsymbol{Y} & \boldsymbol{U} = CX_{10}(\boldsymbol{I} \otimes \boldsymbol{HS}^{\dagger}) \\
\boldsymbol{Y} \otimes \boldsymbol{X} & \boldsymbol{U} = CX_{10}(\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{H}) \\
\boldsymbol{X} \otimes \boldsymbol{Y} & \boldsymbol{U} = CX_{10}(\boldsymbol{H} \otimes \boldsymbol{HS}^{\dagger}) \\
\boldsymbol{Y} \otimes \boldsymbol{Y} & \boldsymbol{U} = CX_{10}(\boldsymbol{HS}^{\dagger} \otimes \boldsymbol{HS}^{\dagger})
\end{array}
$$

Additional remarks

Here, the CNOT (CX10) operation appears for the following reason. Each Pauli measurement that does not include the identity matrix is equivalent up to a unitary to $\boldsymbol{Z} \otimes \boldsymbol{Z}$. The eigenvalues $\boldsymbol{Z} \otimes \boldsymbol{Z}$ of only depend on the parity of the qubits that comprise each computational basis vector, and the controlled-not operations serve to compute this parity and store it in the first bit. Then once the first bit is measured, you can recover the identity of the resultant half-space, which is equivalent to measuring the Pauli operator.

Reducing space

Also, while it can be tempting to assume that measuring $\boldsymbol{Z} \otimes \boldsymbol{Z}$ is the same as sequentially measuring $\boldsymbol{Z} \otimes \boldsymbol{I}$ and then $\boldsymbol{I} \otimes \boldsymbol{Z}$, this assumption would be false. The reason is that measuring $\boldsymbol{Z} \otimes \boldsymbol{Z}$ projects the quantum state into either the $+1$ or -1 eigenstate of these operators. Measuring $\boldsymbol{Z} \otimes \boldsymbol{I}$ and then $\boldsymbol{I} \otimes \boldsymbol{Z}$ projects the quantum state vector first onto a half space of $\boldsymbol{Z} \otimes \boldsymbol{I}$ and then onto a half space of $\boldsymbol{I} \otimes \boldsymbol{Z}$. As there are four computational basis vectors, performing both measurements reduces the state to a quarter-space and hence reduces it to a single computational basis vector.

Two-qubit Hamiltonian

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns; sns.set_theme(font_scale=1.5)
from tqdm import tqdm
sigma_x = np.array([[0, 1], [1, 0]])
sigma_z = np.array([[1, 0], [0, -1]])
ket0 = np.array([1, 0])
ket1 = np.array([0, 1])
I = np.eye(2)

def Hamiltonian(lmb):
    Hx = 2.0
    Hz = 3.0
    H0Energiesnoninteracting = [0.0, 2.5, 6.5, 7.0]

    HI = Hz*np.kron(sigma_z, sigma_z) + Hx*np.kron(sigma_x, sigma_x)
    H0 = np.diag(H0Energiesnoninteracting)
    H = H0 + lmb*HI
    return H


def trace_out(state, index):
    density = np.outer(state, np.conj(state))
    if index == 0:
        op0 = np.kron(ket0, I)
        op1 = np.kron(ket1, I)
    elif index == 1:
        op0 = np.kron(I, ket0)
```

Lipkin model

We will study a schematic model (the Lipkin model, see Nuclear Physics **62** (1965) 188), for the interaction among 2 and more fermions that can occupy two different energy levels.
In project 1 we consider a two-fermion case and a four-fermion case. For four fermions, the case we consider in the examples here, each levels has degeneration $d = 4$, leading to different total spin values. The two levels have quantum numbers $\sigma = \pm 1$, with the upper level having $2\sigma = +1$ and energy $\varepsilon_1 = \varepsilon/2$. The lower level has $2\sigma = -1$ and energy $\varepsilon_2 = -\varepsilon/2$. That is, the lowest single-particle level has negative spin projection (or spin down), while the upper level has spin up. In addition, the substates of each level are characterized by the quantum numbers $p = 1, 2, 3, 4$.

Four fermion case

We define the single-particle states (for the four fermion case which we will work on here)

$$|u_{\sigma=-1,p}\rangle = a^{\dagger}_{-p}|0\rangle \qquad |u_{\sigma=1,p}\rangle = a^{\dagger}_{+p}|0\rangle.$$

The single-particle states span an orthonormal basis.

Hamiltonian

The Hamiltonian of the system is given by

$$\hat{H} = \quad \hat{H}_0 + \hat{H}_1 + \hat{H}_2$$

$$\hat{H}_0 = \quad \tfrac{1}{2}\varepsilon \textstyle\sum_{\sigma,p} \sigma a^\dagger_{\sigma,p} a_{\sigma,p}$$

$$\hat{H}_1 = \quad \tfrac{1}{2} V \textstyle\sum_{\sigma,p,p'} a^\dagger_{\sigma,p} a^\dagger_{\sigma,p'} a_{-\sigma,p'} a_{-\sigma,p}$$

$$\hat{H}_2 = \quad \tfrac{1}{2} W \textstyle\sum_{\sigma,p,p'} a^\dagger_{\sigma,p} a^\dagger_{-\sigma,p'} a_{\sigma,p'} a_{-\sigma,p}$$

where V and W are constants. The operator H_1 can move pairs of fermions while H_2 is a spin-exchange term. The latter moves a pair of fermions from a state $(p\sigma, p'-\sigma)$ to a state $(p-\sigma, p'\sigma)$.

Quasispin operators

We are going to rewrite the above Hamiltonian in terms of so-called quasispin operators

$$\hat{J}_+ = \sum_p a^\dagger_{p+} a_{p-}$$

$$\hat{J}_- = \sum_p a^\dagger_{p-} a_{p+}$$

$$\hat{J}_z = \frac{1}{2} \sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}$$

$$\hat{J}^2 = J_+ J_- + J_z^2 - J_z$$

We show here that these operators obey the commutation relations for angular momentum.

Including the number operator

We can in turn express $\hat{H}$ in terms of the above quasispin operators and the number operator

$$\hat{N} = \sum_{p\sigma} a^\dagger_{p\sigma} a_{p\sigma}.$$

We have the following quasispin operators

$$J_\pm = \sum_p a^\dagger_{p\pm} a_{p\mp}, \tag{1}$$

$$J_z = \frac{1}{2}\sum_{p,\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}, \tag{2}$$

$$J^2 = J_+ J_- + J_z^2 - J_z, \tag{3}$$

and we want to compute the commutators

$$[J_z, J_\pm], \quad [J_+, J_-], \quad [J^2, J_\pm] \quad \text{og} \quad [J^2, J_z].$$

Angular momentum magics I

Let us start with the first one and inserting for J_z and $J_\pm$ given by the equations (2) and (1), respectively, we obtain

$$
\begin{aligned}
[J_z, J_\pm] &= J_z J_\pm - J_\pm J_z \\
&= \left(\frac{1}{2}\sum_{p,\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}\right)\left(\sum_{p'} a^\dagger_{p'\pm} a_{p'\mp}\right) - \left(\sum_{p'} a^\dagger_{p'\pm} a_{p'\mp}\right)\left(\frac{1}{2}\sum_{p,\sigma}\right. \\
&= \frac{1}{2}\sum_{p,p',\sigma} \sigma \left(a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm} a_{p'\mp} a^\dagger_{p\sigma} a_{p\sigma}\right).
\end{aligned}
$$

Angular momentum magics II

Using the commutation relations for the creation and annihilation operators

$$\{a_l, a_k\} = 0, \tag{4}$$
$$\{a_l^\dagger, a_k^\dagger\} = 0, \tag{5}$$
$$\{a_l^\dagger, a_k\} = \delta_{lk}, \tag{6}$$

in order to move the operators in the right product to be in the same order as those in the lefthand product

$$[J_z, J_\pm] = \frac{1}{2} \sum_{p,p',\sigma} \sigma \left(a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm} \left(\delta_{p'p} \delta_{\mp\sigma} - a^\dagger_{p\sigma} a_{p'\mp} \right) a_{p\sigma} \right)$$
$$= \frac{1}{2} \sum_{p,p',\sigma} \sigma \left(a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm} \delta_{p'p} \delta_{\mp\sigma} a_{p\sigma} + a^\dagger_{p'\pm} a^\dagger_{p\sigma} a_{p'\mp} a_{p\sigma} \right)$$

Angular momentum magics III

It results in

$$
\begin{aligned}
[J_z, J_\pm] &= \frac{1}{2} \sum_{p,p',\sigma} \sigma \Big(a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm} \delta_{pp'} \delta_{\mp\sigma} a_{p\sigma} + a^\dagger_{p\sigma} a^\dagger_{p'\pm} a_{p\sigma} a_{p'\mp} \\
&= \frac{1}{2} \sum_{p,p',\sigma} \sigma \Big(a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm} \delta_{pp'} \delta_{\mp\sigma} a_{p\sigma} + a^\dagger_{p\sigma} \Big(\delta_{pp'} \delta_{\pm\sigma} - \\
&= \frac{1}{2} \sum_{p,p',\sigma} \sigma \Big(a^\dagger_{p\sigma} \delta_{pp'} \delta_{\pm\sigma} a_{p'\mp} - a^\dagger_{p'\pm} \delta_{pp'} \delta_{\mp\sigma} a_{p\sigma} \Big) .
\end{aligned}
$$

Angular momentum magics IV

The last equality leads to

$$[J_z, J_\pm] = \frac{1}{2}\sum_p \left((\pm 1)a^\dagger_{p\pm}a_{p\mp} - (\mp 1)a^\dagger_{p\pm}a_{p\mp}\right) = \pm\frac{1}{2}\sum_p \left(a^\dagger_{p\pm}a_{p\mp} + (\pm 1$$

$$= \pm\sum_p a^\dagger_{p\pm}a_{p\mp} = \pm J_\pm,$$

where the last results follows from comparing with Eq. (1).

Angular momentum magics V

We can then continue with the next commutation relation, using Eq. (1),

$$
\begin{aligned}
[J_+, J_-] &= J_+J_- - J_-J_+ \\
&= \left(\sum_p a^\dagger_{p'+}a_{p-}\right)\left(\sum_{p'} a^\dagger_{p'-}a_{p'+}\right) - \left(\sum_{p'} a^\dagger_{p'-}a_{p'+}\right)\left(\sum_p a^\dagger_{p+}\right. \\
&= \sum_{p,p'}\left(a^\dagger_{p'+}a_{p-}a^\dagger_{p'-}a_{p'+} - a^\dagger_{p'-}a_{p'+}a^\dagger_{p+}a_{p-}\right) \\
&= \sum_{p,p'}\left(a^\dagger_{p'+}a_{p-}a^\dagger_{p'-}a_{p'+} - a^\dagger_{p'-}\left(\delta_{++}\delta_{pp'} - a^\dagger_{p+}a_{p'+}\right)a_{p-}\right) \\
&= \sum_{p,p'}\left(a^\dagger_{p'+}a_{p-}a^\dagger_{p'-}a_{p'+} - a^\dagger_{p'-}\delta_{pp'}a_{p-} + a^\dagger_{p'-}a^\dagger_{p+}a_{p'+}a_{p-}\right) \\
&= \sum_{p,p'}\left(a^\dagger_{p'+}a_{p-}a^\dagger_{p'-}a_{p'+} - a^\dagger_{p'-}\delta_{pp'}a_{p-} + a^\dagger_{p+}a^\dagger_{p'-}a_{p-}a_{p'+}\right) \\
&= \sum_{p,p'}\left(a^\dagger_{p'+}a_{p-}a^\dagger_{p'-}a_{p'+} - a^\dagger_{p'-}\delta_{pp'}a_{p-} + a^\dagger_{p+}\left(\delta_{--}\delta_{pp'} - a_{p-}a^\dagger_{p'}\right.\right.
\end{aligned}
$$

Angular momentum magics VI

Which results in

$$[J_+, J_-] = \sum_p \left(a^\dagger_{p+} a_{p+} - a^\dagger_{p-} a_{p-} \right) = 2J_z,$$

It is straightforward to show that

$$[J^2, J_\pm] = [J_+ J_- + J_z^2 - J_z, J_\pm] = [J_+ J_-, J_\pm] + [J_z^2, J_\pm] - [J_z, J_\pm].$$

Angular momentum magics VII

Using the relations

$$[AB, C] = A[B, C] + [A, C]B, \tag{7}$$

$$[A, BC] = [A, B]C + B[A, C], \tag{8}$$

we obtain

$$[J^2, J_\pm] = J_+[J_-, J_\pm] + [J_+, J_\pm]J_- + J_z[J_z, J_\pm] + [J_z, J_\pm]J_z - [J_z, J_\pm].$$

Angular momentum magics VIII

Finally, from the above it follows that

$$\begin{aligned}[J^2, J_+] &= -2J_+J_z + J_z[J_z, J_+] + [J_z, J_+]J_z - [J_z, J_+] \\ &= -2J_+J_z + J_zJ_+ + J_+J_z - J_+ \\ &= -2J_+J_z + J_+ + J_+J_z + J_+J_z - J_+ = 0,\end{aligned}$$

and

$$\begin{aligned}[J^2, J_-] &= 2J_zJ_- + J_z[J_z, J_-] + [J_z, J_-]J_z - [J_z, J_-] \\ &= 2J_zJ_- - J_zJ_- - J_-J_z + J_- \\ &= J_zJ_- - (J_zJ_- + J_-) + J_- = 0.\end{aligned}$$

Angular momentum magics IX

Our last commutator is given by

$$
\begin{aligned}
[J^2, J_z] &= [J_+J_- + J_z^2 - J_z, J_z] \\
&= [J_+J_-, J_z] + [J_z^2, J_z] - [J_z, J_z] \\
&= J_+[J_-, J_z] + [J_+, J_z]J_- \\
&= J_+J_- - J_+J_- = 0
\end{aligned}
$$

Angular momentum magics X

Summing up we have

$$[J_z, J_\pm] = \pm J_\pm, \tag{9}$$
$$[J_+, J_-] = 2J_z, \tag{10}$$
$$[J^2, J_\pm] = 0, \tag{11}$$
$$[J^2, J_z] = 0, \tag{12}$$

which are the standard commutation relations for angular (or orbital) momentum $L_\pm$, L_z og L^2.

Rewriting the Hamiltonian

We wrote the above Hamiltonian as

$$H = H_0 + H_1 + H_2,$$

with

$$H_0 = \frac{1}{2}\varepsilon \sum_{p\sigma} \sigma a^{\dagger}_{p\sigma} a_{p\sigma},$$

and

$$H_1 = \frac{1}{2}V \sum_{p,p',\sigma} a^{\dagger}_{p\sigma} a^{\dagger}_{p'\sigma} a_{p'-\sigma} a_{p-\sigma},$$

and

$$H_2 = \frac{1}{2}W \sum_{p,p',\sigma} a^{\dagger}_{p\sigma} a^{\dagger}_{p'-\sigma} a_{p'\sigma} a_{p-\sigma}.$$

Hamiltonian and angular momentum magics I

We will now rewrite the Hamiltonian in terms of the above quasi-spin operators and the number operator

$$N = \sum_{p,\sigma} a^{\dagger}_{p\sigma} a_{p\sigma}. \tag{13}$$

Going through each term of the Hamiltonian and using the expressions for the quasi-spin operators we obtain

$$H_0 = \varepsilon J_z. \tag{14}$$

Hamiltonian and angular momentum magics II

Moving over to H_1 and using the anti-commutation relations (4) through (6) we obtain

$$
\begin{aligned}
H_1 &= \frac{1}{2} V \sum_{p,p',\sigma} a^{\dagger}_{p\sigma} a^{\dagger}_{p'\sigma} a_{p'-\sigma} a_{p-\sigma} \\
&= \frac{1}{2} V \sum_{p,p',\sigma} -a^{\dagger}_{p\sigma} a^{\dagger}_{p'\sigma} a_{p-\sigma} a_{p'-\sigma} \\
&= \frac{1}{2} V \sum_{p,p',\sigma} -a^{\dagger}_{p\sigma} \left(\delta_{pp'} \delta_{\sigma-\sigma} - a_{p-\sigma} a^{\dagger}_{p'\sigma} \right) a_{p'-\sigma} \\
&= \frac{1}{2} V \sum_{p,p',\sigma} a^{\dagger}_{p\sigma} a_{p-\sigma} a^{\dagger}_{p'\sigma} a_{p'-\sigma}
\end{aligned}
$$

Hamiltonian and angular momentum magics III

Rewriting the sum over σ we arrive at

$$
\begin{aligned}
H_1 &= \frac{1}{2} V \sum_{p,p'} a^\dagger_{p+} a_{p-} a^\dagger_{p'+} a_{p'-} + a^\dagger_{p-} a_{p+} a^\dagger_{p'-} a_{p'+} \\
&= \frac{1}{2} V \Bigg[\sum_{p} \left(a^\dagger_{p+} a_{p-} \right) \sum_{p'} \left(a^\dagger_{p'+} a_{p'-} \right) + \sum_{p} \left(a^\dagger_{p-} a_{p+} \right) \sum_{p'} \left(a^\dagger_{p'-} a_{p'+} \right) \\
&= \frac{1}{2} V \left[J_+ J_+ + J_- J_- \right] = \frac{1}{2} V \left[J_+^2 + J_-^2 \right],
\end{aligned}
$$

which leads to

$$
H_1 = \frac{1}{2} V \left(J_+^2 + J_-^2 \right). \tag{15}
$$

Hamiltonian and angular momentum magics IV

Finally, we rewrite the last term

$$
\begin{aligned}
H_2 &= \frac{1}{2} W \sum_{p,p',\sigma} a^\dagger_{p\sigma} a^\dagger_{p'-\sigma} a_{p'\sigma} a_{p-\sigma} \\
&= \frac{1}{2} W \sum_{p,p',\sigma} -a^\dagger_{p\sigma} a^\dagger_{p'-\sigma} a_{p-\sigma} a_{p'\sigma} \\
&= \frac{1}{2} W \sum_{p,p',\sigma} -a^\dagger_{p\sigma} \left(\delta_{pp'} \delta_{-\sigma-\sigma} - a_{p-\sigma} a^\dagger_{p'-\sigma} \right) a_{p'\sigma} \\
&= \frac{1}{2} W \sum_{p,p',\sigma} -a^\dagger_{p\sigma} \delta_{pp'} a_{p'\sigma} + a^\dagger_{p\sigma} a_{p-\sigma} a^\dagger_{p'-\sigma} a_{p'\sigma} \\
&= \frac{1}{2} W \left(-\sum_{p,\sigma} a^\dagger_{p\sigma} a_{p\sigma} + \sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p-\sigma} a^\dagger_{p'-\sigma} a_{p'\sigma} \right)
\end{aligned}
$$

Hamiltonian and angular momentum magics V

Using the expression for the number operator we obtain

$$
\begin{aligned}
\sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p-\sigma} a^\dagger_{p'-\sigma} a_{p'\sigma} &= \sum_{p,p'} a^\dagger_{p+} a_{p-} a^\dagger_{p'-} a_{p'+} + a^\dagger_{p-} a_{p+} a^\dagger_{p'+} a_{p'-} \\
&= \sum_{p} \left(a^\dagger_{p+} a_{p-} \right) \sum_{p'} \left(a^\dagger_{p'-} a_{p'+} \right) + \sum_{p} \left(a^\dagger_{p-} a_{p+} \right) \\
&= J_+ J_- + J_- J_+,
\end{aligned}
$$

resulting in

$$
H_2 = \frac{1}{2} W \left(-N + J_+ J_- + J_- J_+ \right). \tag{16}
$$

We have thus expressed the Hamiltonian in term of the quasi-spin operators. Below, we will show how we can rewrite these expressions in terms of Pauli X, Y and Z matrices.

Commutation relations for the Hamiltonian

The above expressions can in turn be used to show that the Hamiltonian commutes with the various quasi-spin operators. This leads to quantum numbers which are conserved. Let us first show that $[H, J^2] = 0$, which means that J is a so-called *good* quantum number and that the total spin is a conserved quantum number. We have

$$
\begin{aligned}
[H, J^2] &= [H_0 + H_1 + H_2, J^2] \\
&= [H_0, J^2] + [H_1, J^2] + [H_2, J^2] \\
&= \varepsilon [J_z, J^2] + \frac{1}{2} V [J_+^2 + J_-^2, J^2] + \frac{1}{2} W [-N + J_+ J_- + J_- J_+, J^2].
\end{aligned}
$$

Hamiltonian and commutators

We have previously shown that

$$[H, J^2] = \frac{1}{2}V\left([J_+^2, J^2] + [J_-^2, J^2]\right) + \frac{1}{2}W\left(-[N, J^2] + [J_+J_-, J^2] + [J_-J_+,\right.$$

Using that $[J_\pm, J^2] = 0$, it follows that $[J_\pm^2, J^2] = 0$. We can then see that $[J_+J_-, J^2] = 0$ and $[J_-J_+, J^2] = 0$ which leads to

$$\begin{aligned}
[H, J^2] &= -\frac{1}{2}W[N, J^2] \\
&= \frac{1}{2}W\left(-[N, J_+J_-] - [N, J_z^2] + [N, J_z]\right) \\
&= \frac{1}{2}W\left(-[N, J_+]J_- - J_+[N, J_-] - [N, J_z]J_z - J_z[N, J_z] + [N, J_z]\right)
\end{aligned}$$

Including the number operator

Combining with the number operator we have

$$
\begin{aligned}
[N, J_\pm] &= NJ_\pm - J_\pm N \\
&= \left(\sum_{p,\sigma} a^\dagger_{p\sigma} a_{p\sigma}\right)\left(\sum_{p'} a^\dagger_{p'\pm} a_{p'\mp}\right) - \left(\sum_{p'} a^\dagger_{p'\pm} a_{p'\mp}\right)\left(\sum_{p,\sigma} a^\dagger_{p\sigma} a_{p\sigma}\right) \\
&= \sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm} a_{p'\mp} a^\dagger_{p\sigma} a_{p\sigma} \\
&= \sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm}\left(\delta_{\mp\sigma}\delta_{pp'} - a^\dagger_{p\sigma} a_{p'\mp}\right) a_{p\sigma} \\
&= \sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} - a^\dagger_{p'\pm}\delta_{\mp\sigma}\delta_{pp'} a_{p\sigma} + a^\dagger_{p'\pm} a^\dagger_{p\sigma} a_{p'\mp} a_{p\sigma} \\
&= \sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} + a^\dagger_{p\sigma} a^\dagger_{p'\pm} a_{p\sigma} a_{p'\mp} - \sum_{p} a^\dagger_{p\pm} a_{p\mp} \\
&= \sum_{p,p',\sigma} a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\pm} a_{p'\mp} + a^\dagger_{p\sigma}\left(\delta_{pp'}\delta_{\pm\sigma} - a_{p\sigma} a^\dagger_{p'\pm}\right) a_{p'\mp} - \sum_{p} a^\dagger_{p\pm} \\
&= \sum a^\dagger_{p\pm} a_{p\mp} - \sum a^\dagger_{p\pm} a_{p\mp} = 0.
\end{aligned}
$$

Hamiltonian and angular momentum commutators

We obtain then

$$
\begin{aligned}
[N, J_z] &= NJ_z - J_z N \\
&= \left(\sum_{p,\sigma} a^\dagger_{p\sigma} a_{p\sigma}\right)\left(\frac{1}{2}\sum_{p',\sigma} \sigma a^\dagger_{p'\sigma} a_{p'\sigma}\right) - \left(\frac{1}{2}\sum_{p',\sigma} \sigma a^\dagger_{p'\sigma} a_{p'\sigma}\right)\left(\sum_{p,\sigma} a\right. \\
&= \sum_{p,p',\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_{p'\sigma} a_{p'\sigma} - \sigma a^\dagger_{p'\sigma} a_{p'\sigma} a^\dagger_{p\sigma} a_{p\sigma} = 0,
\end{aligned}
$$

which leads to

$$
[H, J^2] = 0, \tag{17}
$$

and J is a good quantum number.

Constructing the Hamiltonian matrix for $J = 2$

We start with the state (unique) where all spins point down

$$|2, -2\rangle = a^{\dagger}_{1-} a^{\dagger}_{2-} a^{\dagger}_{3-} a^{\dagger}_{4-} |0\rangle \tag{18}$$

which is a state with $J_z = -2$ and $J = 2$. (we label the states as $|J, J_z\rangle$). For $J = 2$ we have the spin projections $J_z = -2, -1, 0, 1, 2$. We can use the lowering and raising operators for spin in order to define the other states

$$J_{+}|J, J_z\rangle = \sqrt{J(J+1) - J_z(J_z+1)}|J, J_z + 1\rangle, \tag{19}$$

$$J_{-}|J, J_z\rangle = \sqrt{J(J+1) - J_z(J_z-1)}|J, J_z - 1\rangle. \tag{20}$$

Constructing the Hamiltonian matrix for the other states

We can then construct all other states with $J = 2$ using the raising operator J_+ on $|2,-2\rangle$

$$J_+|2,-2\rangle = \sqrt{2(2+1)-(-2)(-2+1)}|2,-2+1\rangle = \sqrt{6-2}|2,-1\rangle = 2|2$$

which gives

$$\begin{aligned}|2,-1\rangle &= \frac{1}{2}J_+|2,-2\rangle \\ &= \frac{1}{2}\sum_p a^\dagger_{p+}a_{p-}a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4-}|0\rangle \\ &= \frac{1}{2}\left(a^\dagger_{1+}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2+}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3+}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2-}\right.\end{aligned} \quad (21)$$

Constructing the Hamiltonian matrix

We can construct all the other states in the same way. That is

$$J_+|2,-1\rangle = \sqrt{2(2+1)-(-1)(-1+1)}|2,-1+1\rangle = \sqrt{6}|2,0\rangle,$$

which results in

$$|2,0\rangle = \frac{1}{\sqrt{6}}\left(a^\dagger_{1+}a^\dagger_{2+}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1+}a^\dagger_{2-}a^\dagger_{3+}a^\dagger_{4-} + a^\dagger_{1+}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4+} + a^\dagger_{1-}a^\dagger_{2+}\right.$$
$$\left. a^\dagger_{1-}a^\dagger_{2+}a^\dagger_{3-}a^\dagger_{4+} + a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3+}a^\dagger_{4+}\right)|0\rangle \tag{22}$$

Constructing the Hamiltonian matrix, last two states

The two remaining states are

$$|2,1\rangle = \frac{1}{2}\left(a^{\dagger}_{1+}a^{\dagger}_{2+}a^{\dagger}_{3+}a^{\dagger}_{4-} + a^{\dagger}_{1+}a^{\dagger}_{2+}a^{\dagger}_{3-}a^{\dagger}_{4+} + a^{\dagger}_{1+}a^{\dagger}_{2-}a^{\dagger}_{3+}a^{\dagger}_{4+} + a^{\dagger}_{1-}a^{\dagger}_{2+}a^{\dagger}_{3} \tag{23}$$

and

$$|2,2\rangle = a^{\dagger}_{1+}a^{\dagger}_{2+}a^{\dagger}_{3+}a^{\dagger}_{4+}|0\rangle. \tag{24}$$

Final Hamiltonian matrix

These five states can in turn be used as computational basis states in order to define the Hamiltonian matrix to be diagonalized. The matrix elements are given by $\langle J, J_z|H|J', J_z'\rangle$. The Hamiltonian is hermitian and we obtain after all this labor of ours

$$H_{J=2} = \begin{bmatrix} -2\varepsilon & 0 & \sqrt{6}V & 0 & 0 \\ 0 & -\varepsilon + 3W & 0 & 3V & 0 \\ \sqrt{6}V & 0 & 4W & 0 & \sqrt{6}V \\ 0 & 3V & 0 & \varepsilon + 3W & 0 \\ 0 & 0 & \sqrt{6}V & 0 & 2\varepsilon \end{bmatrix} \qquad (25)$$

Comparing with standard diagonalization

We can now select a set of parameters and diagonalize the above matrix. We select $\epsilon = 2$, $V = -1/3$, $W = -1/4$ and our matrix becoes

$$H_{J=2}^{(1)} = \begin{bmatrix} -4 & 0 & -\sqrt{6}/3 & 0 & 0 \\ 0 & -2-3/4 & 0 & -1 & 0 \\ -\sqrt{6}/3 & 0 & -1 & 0 & -\sqrt{6}/3 \\ 0 & -1 & 0 & 2+-3/4 & 0 \\ 0 & 0 & -\sqrt{6}/3 & 0 & 4 \end{bmatrix},$$

which gives the eigenvalue

$$D = \begin{bmatrix} -4.21288 & 0 & 0 & 0 & 0 \\ 0 & -2.98607 & 0 & 0 & 0 \\ 0 & 0 & -0.91914 & 0 & 0 \\ 0 & 0 & 0 & 1.48607 & 0 \\ 0 & 0 & 0 & 0 & 4.13201 \end{bmatrix}.$$

The lowest state has an admixture of basis states given by

$$|\psi_0\rangle = 0.96735|2,-2\rangle + 0.25221|2,0\rangle + 0.02507|2,2\rangle,$$

with energy $E_0 = -4.21288$

Comparing with standard diagonalization, other parameters

We can now change the parameters to $\varepsilon = 2$, $V = -4/3$, $W = -1$. Our matrix reads then

$$H^{(2)}_{J=2} = \begin{bmatrix} -4 & 0 & -4\sqrt{6}/3 & 0 & 0 \\ 0 & -5 & 0 & -4 & 0 \\ -4\sqrt{6}/3 & 0 & -4 & 0 & -4\sqrt{6}/3 \\ 0 & -4 & 0 & -1 & 0 \\ 0 & 0 & -4\sqrt{6}/3 & 0 & 4 \end{bmatrix},$$

with the following eigenvalues

$$D = \begin{bmatrix} -7.75122 & 0 & 0 & 0 & 0 \\ 0 & -7.47214 & 0 & 0 & 0 \\ 0 & 0 & -1.55581 & 0 & 0 \\ 0 & 0 & 0 & 1.47214 & 0 \\ 0 & 0 & 0 & 0 & 5.30704 \end{bmatrix}.$$

The new ground state (lowest state) has the following admixture of computational basis states

$$|\psi_0\rangle = 0.64268|2,-2\rangle + 0.73816|2,0\rangle + 0.20515|2,2\rangle,$$

with energy $E_0 = -7.75122$.

Analysis

For the first set of parameters, the likelihood for observing the system in the computational basis state $|2, -2\rangle$ is rather large. This is expected since the interaction matrix elements are smaller than the single-particle energies. For the second case, with larger matrix elements, we see a much stronger mixing of the other states, again as expected due to the ratio of the interaction matrix elements and the single-particle energies.

Quantum computing and solving the eigenvalue problem for the Lipkin model

We turn now to a simpler variant of the Lipkin model without the W-term and a total spin of $J = 1$ only as maximum value of the spin. This corresponds to a system with $N = 2$ particles (fermions in our case). Our Hamiltonian is given by the quasispin operators (see below)

$$\hat{H} = \epsilon \hat{J}_z - \frac{1}{2} V(\hat{J}_+ \hat{J}_+ + \hat{J}_- \hat{J}_-).$$

As discussed earlier the quasispin operators act like lowering and raising angular momentum operators.

With these properties we can calculate the Hamiltonian matrix for the Lipkin model by computing the various matrix elements

$$\langle JJ_z | H | JJ_z' \rangle, \tag{26}$$

where the non-zero elements are given by

$$\begin{aligned} \langle JJ_z | H | JJ_z' \rangle &= \epsilon J_z \\ \langle JJ_z | H | JJ_z' \pm 2 \rangle &= \langle JJ_z \pm 2 | H | JJ_z' \rangle \\ &= -\frac{1}{2} VC \end{aligned}$$

Plans for next week

1. We will focus mainly on setting up the quantum circuit for the Lipkin model with $J = 1$ and $J = 2$, rewriting the Lipkin model in terms of Pauli matrices